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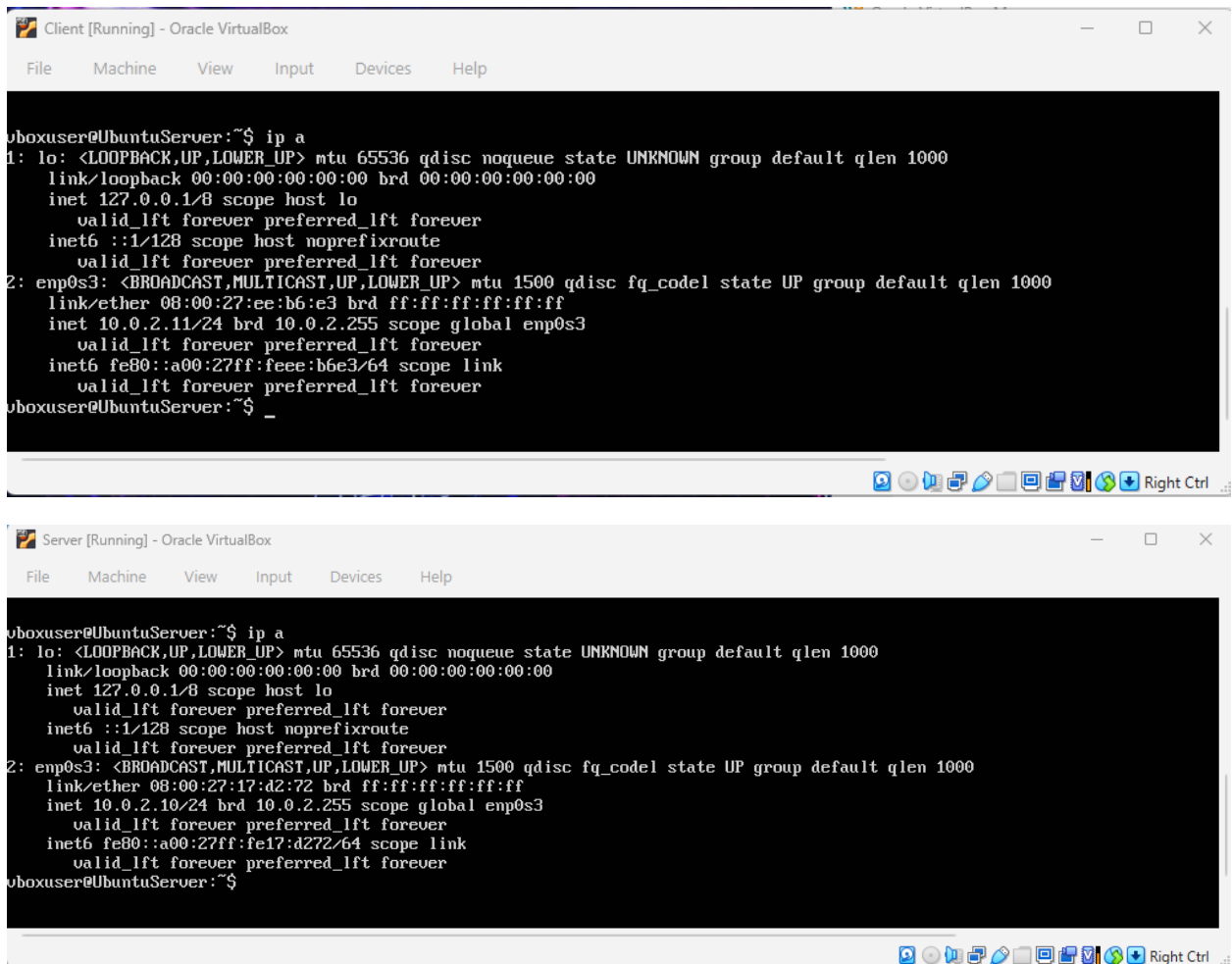
Professor Kaur

Lab 3: Secure Client-Server Communication Using DES Encryption

2/3/2026

Task 1:

Screenshot showing IP addresses of Client and Server VMs

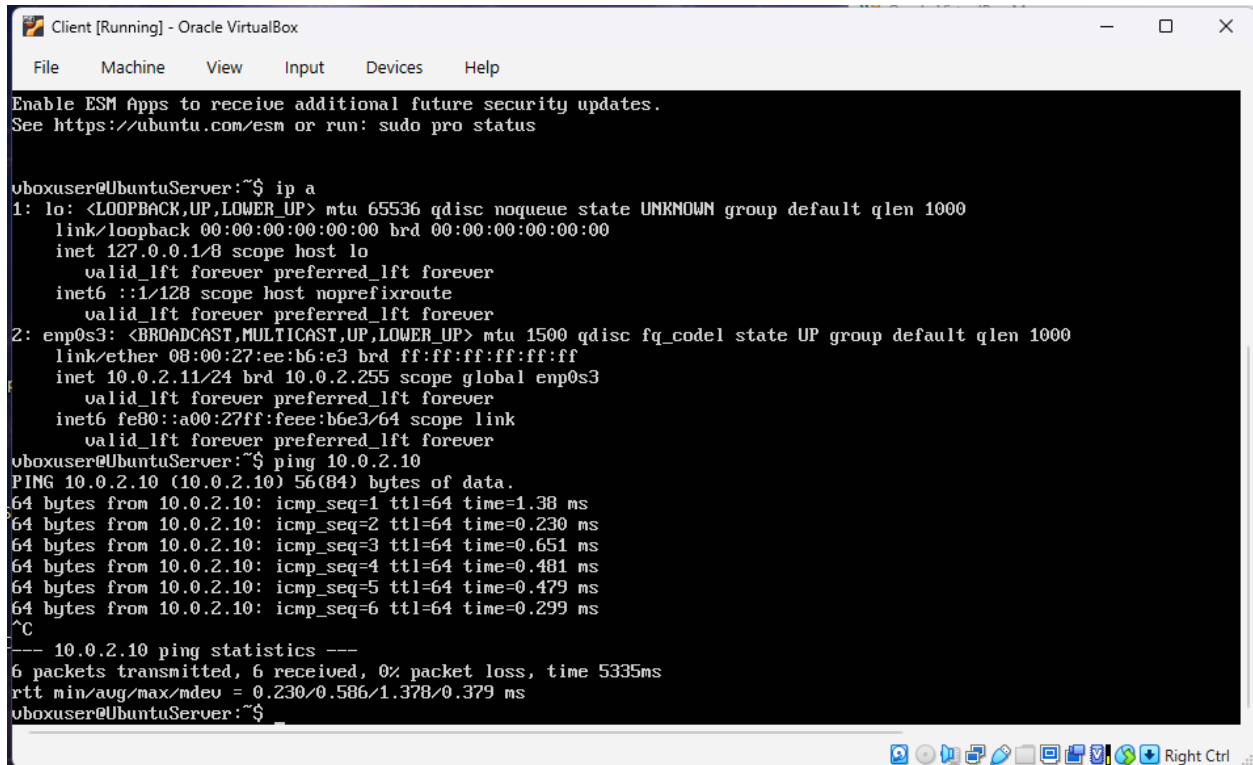


The image displays two screenshots of Oracle VM VirtualBox terminal windows. The top window, titled 'Client [Running] - Oracle VirtualBox', shows the output of the 'ip a' command on a 'vboxuser@UbuntuServer' machine. It lists the loopback interface 'lo' with IP 127.0.0.1 and the ethernet interface 'enp0s3' with IP 10.0.2.11. The bottom window, titled 'Server [Running] - Oracle VirtualBox', shows the output of the 'ip a' command on a 'vboxuser@UbuntuServer' machine. It lists the loopback interface 'lo' with IP 127.0.0.1 and the ethernet interface 'enp0s3' with IP 10.0.2.10. Both windows show the standard output of the 'ip a' command, including details about the interfaces and their configurations.

```
vboxuser@UbuntuServer:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ee:b6:e3 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.11/24 brd 10.0.2.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feee:b6e3/64 scope link
        valid_lft forever preferred_lft forever
vboxuser@UbuntuServer:~$ _
```

```
vboxuser@UbuntuServer:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:17:d2:72 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.10/24 brd 10.0.2.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe17:d272/64 scope link
        valid_lft forever preferred_lft forever
vboxuser@UbuntuServer:~$
```

Screenshot of successful ping from Client to Server



The screenshot shows a terminal window titled "Client [Running] - Oracle VirtualBox". The terminal output displays the results of the 'ip a' command, showing the loopback interface 'lo' and the ethernet interface 'enp0s3'. It then shows a successful ping from '10.0.2.10' to '10.0.2.10' with 6 packets transmitted and 0% packet loss.

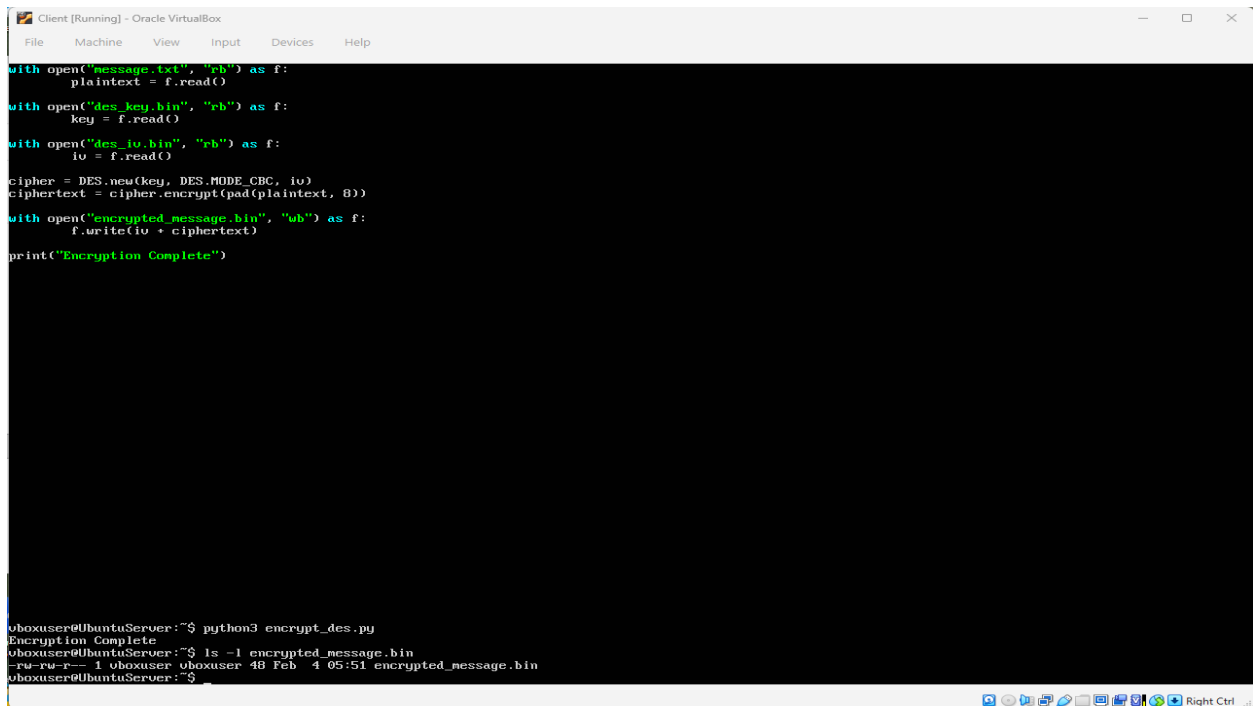
```
Client [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

vboxuser@UbuntuServer:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ee:b6:e3 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.11/24 brd 10.0.2.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feee:b6e3/64 scope link
        valid_lft forever preferred_lft forever
vboxuser@UbuntuServer:~$ ping 10.0.2.10
PING 10.0.2.10 (10.0.2.10) 56(84) bytes of data.
64 bytes from 10.0.2.10: icmp_seq=1 ttl=64 time=1.38 ms
64 bytes from 10.0.2.10: icmp_seq=2 ttl=64 time=0.230 ms
64 bytes from 10.0.2.10: icmp_seq=3 ttl=64 time=0.651 ms
64 bytes from 10.0.2.10: icmp_seq=4 ttl=64 time=0.481 ms
64 bytes from 10.0.2.10: icmp_seq=5 ttl=64 time=0.479 ms
64 bytes from 10.0.2.10: icmp_seq=6 ttl=64 time=0.299 ms
^C
--- 10.0.2.10 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5335ms
rtt min/avg/max/mdev = 0.230/0.586/1.378/0.379 ms
vboxuser@UbuntuServer:~$
```

Task 2:

Screenshot of the encrypted file on the sender VM



```
Client [Running] - Oracle VirtualBox
File Machine View Input Devices Help

with open("message.txt", "rb") as f:
    plaintext = f.read()

with open("des_key.bin", "rb") as f:
    key = f.read()

with open("des_iv.bin", "rb") as f:
    iv = f.read()

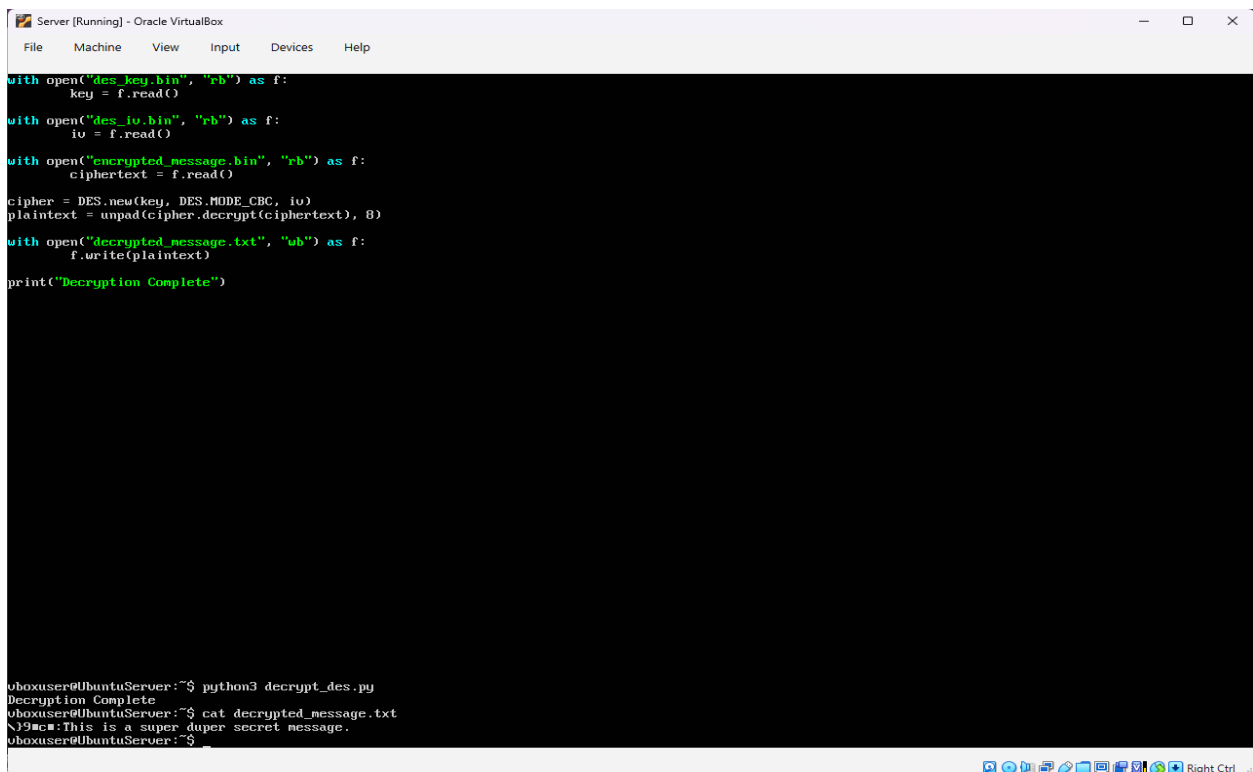
cipher = DES.new(key, DES.MODE_CBC, iv)
ciphertext = cipher.encrypt(pad(plaintext, 8))

with open("encrypted_message.bin", "wb") as f:
    f.write(iv + ciphertext)

print("Encryption Complete")

vboxuser@UbuntuServer:~$ python3 encrypt_des.py
Encryption Complete
vboxuser@UbuntuServer:~$ ls -l encrypted_message.bin
-rw-r--r-- 1 vboxuser vboxuser 48 Feb  4 05:51 encrypted_message.bin
vboxuser@UbuntuServer:~$
```

Screenshot of the decrypted plaintext on the receiver VM.



```
Server [Running] - Oracle VirtualBox
File Machine View Input Devices Help

with open("des_key.bin", "rb") as f:
    key = f.read()

with open("des_iv.bin", "rb") as f:
    iv = f.read()

with open("encrypted_message.bin", "rb") as f:
    ciphertext = f.read()

cipher = DES.new(key, DES.MODE_CBC, iv)
plaintext = unpad(cipher.decrypt(ciphertext), 8)

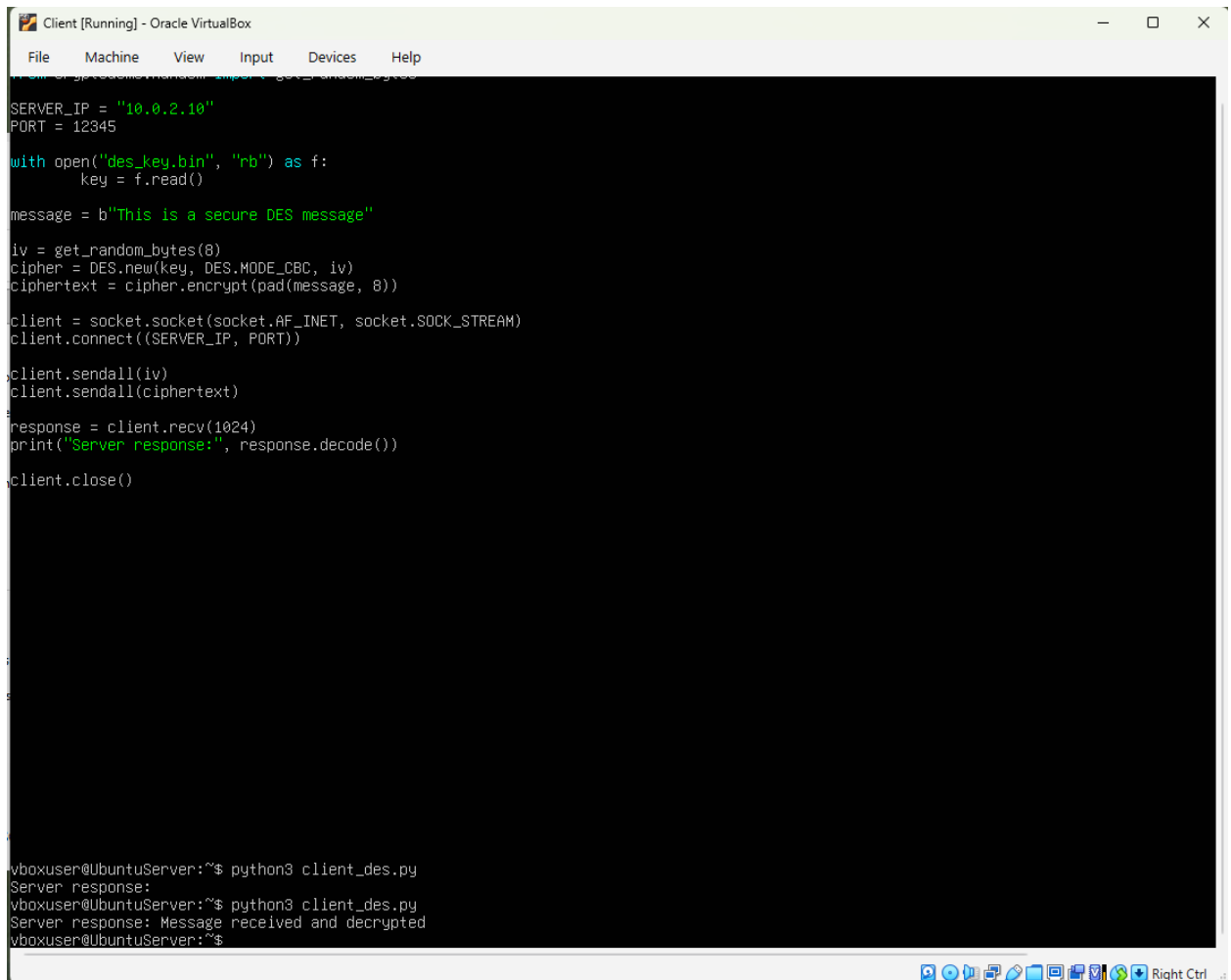
with open("decrypted_message.txt", "wb") as f:
    f.write(plaintext)

print("Decryption Complete")

vboxuser@UbuntuServer:~$ python3 decrypt_des.py
Decryption Complete
vboxuser@UbuntuServer:~$ cat decrypted_message.txt
^J^ca:This is a super duper secret message.
vboxuser@UbuntuServer:~$
```

Task 3:

Screenshot of encrypted message sent by the client.



The screenshot shows a terminal window titled "Client [Running] - Oracle VirtualBox". The terminal displays a Python script for DES encryption and the output of its execution. The script defines a server IP and port, reads a key from a file, encrypts a message, and sends it to the server. The server response is "Message received and decrypted".

```
Client [Running] - Oracle VirtualBox
File Machine View Input Devices Help

SERVER_IP = "10.0.2.10"
PORT = 12345

with open("des_key.bin", "rb") as f:
    key = f.read()

message = b"This is a secure DES message"

iv = get_random_bytes(8)
cipher = DES.new(key, DES.MODE_CBC, iv)
ciphertext = cipher.encrypt(pad(message, 8))

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((SERVER_IP, PORT))

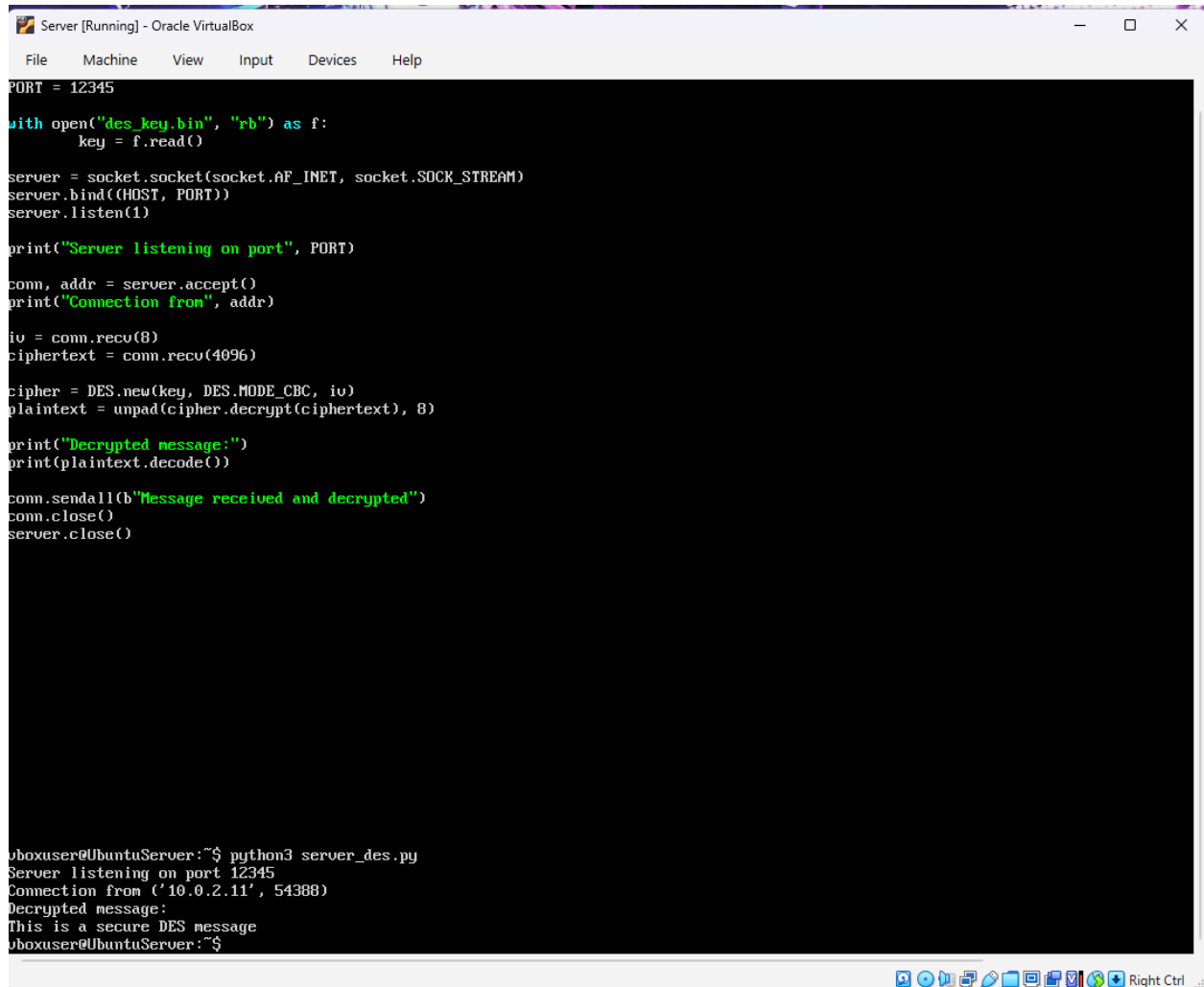
client.sendall(iv)
client.sendall(ciphertext)

response = client.recv(1024)
print('Server response:', response.decode())

client.close()

vboxuser@UbuntuServer:~$ python3 client_des.py
Server response:
vboxuser@UbuntuServer:~$ python3 client_des.py
Server response: Message received and decrypted
vboxuser@UbuntuServer:~$
```

Screenshot of decrypted plaintext on the server.



The screenshot shows a terminal window titled "Server [Running] - Oracle VirtualBox". The terminal displays a Python script for a server that listens on port 12345, accepts a connection, receives ciphertext, decrypts it using DES, and prints the plaintext. The output shows the server successfully decrypting a message.

```
PORT = 12345

with open("des_key.bin", "rb") as f:
    key = f.read()

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((HOST, PORT))
server.listen(1)

print("Server listening on port", PORT)

conn, addr = server.accept()
print("Connection from", addr)

iv = conn.recv(8)
ciphertext = conn.recv(4096)

cipher = DES.new(key, DES.MODE_CBC, iv)
plaintext = unpad(cipher.decrypt(ciphertext), 8)

print("Decrypted message:")
print(plaintext.decode())

conn.sendall(b"Message received and decrypted")
conn.close()
server.close()
```

```
vboxuser@UbuntuServer:~$ python3 server_des.py
Server listening on port 12345
Connection from ('10.0.2.11', 54388)
Decrypted message:
This is a secure DES message
vboxuser@UbuntuServer:~$
```